

Table 1. Mortality and morbidity cost estimates of smoke-induced (health) impacts in studies since 2006
COI, Cost of Illness; WTP, Willingness to Pay; HA, hospital admission; ED, emergency department; VSL Value of Statistical Life;

Author(s)	Mortality cost estimate	Derivation of mortality cost estimate (in 2019 US dollar values)	Morbidity cost estimates	Derivation of morbidity cost estimates
Rittmaster <i>et al.</i> (2006)	VSL estimate: Low: CAN \$3.7 million Central: CAN \$6.3 million High: CAN \$12.7 million	From Environment Canada and Health Canada Air Quality Valuation Model (AQVM) AQVM is a dataset that contains data similar to <i>BenMAP-CE</i> . This includes economic estimates for different health end points taken from relevant studies (1979–94) in the literature	Respiratory HA (COI) Cardiac HA (COI) ED visit (COI) Restricted day activity (WTP, COI) Asthma symptoms (WTP) Acute respiratory symptoms (WTP) The study did not provide COI/WTP estimates in the manuscript	From Environment Canada and Health Canada AQVM
Kochi <i>et al.</i> (2012)	VSL estimate: US\$1.56 million–US \$15.6 million	US Environmental Protection Agency (EPA) estimates (2005) from the Regulatory Impact Analysis for the Final Clean Air Interstate Rule		
Richardson <i>et al.</i> (2012)			WTP for a decrease in 1 symptom day from wildfire smoke using an air cleaner: US\$100.6 Private COI for 1 symptom day: US \$11.43	WTP derived using survey information applying the Defensive Behaviour Method Private COI derived using Alberini and Krupnick (2000) based on medical expenses and wages lost
Richardson <i>et al.</i> (2013)			WTP : COI = 9 To achieve a reduction in 1 symptom day – WTP using Defensive Behaviour Method: US\$103.52 (US\$1.94–5.25, 90% CI) – Mean WTP using Contingent Valuation Method: US\$141.24 – COI: US\$3.6 (US\$27.15–727.42, 90% CI) – COI (including missed leisure time): US\$20.10 (US\$16.81–23.38, CI 90%)	All data were obtained through a survey and analysed with statistical data analysis. Defensive Behaviour Model is taken from Richardson <i>et al.</i> (2012), differences in WTP are due to rounding
Moeltner <i>et al.</i> (2013)			Average treatment cost in hospitals and ambulatory surgical centres (COI) – Respiratory illness: US\$42 000 – Cardiovascular illness: US\$47 000 Treatment cost per 100 acres burned depending on distance from fire or type of fire: US\$145–560	Treatment cost summary statistics based on actual cost data from hospitals Treatment cost per 100 acres burned derived from regression analysis
Rappold <i>et al.</i> (2014)	VSL value for mortality: US \$11.4 million (US\$0–226 million, 95% CI)	Internally adjusted <i>BenMAP-CE</i> value	Empirical data on ED + HA average cost (COI): – Asthma: US\$9800 – Congestive heart failure: US\$3200 <i>BenMAP-CE</i> economic values for respiratory and cardiovascular end-points, work loss	ED +HA visit costs were obtained from 2008 Nationwide Emergency Department Sample (NEDS) and from Healthcare Cost and Utilisation Project (HCUP) and multiplied by number of additional incidences
Jones <i>et al.</i> (2016)			WTP for a decrease in 1 symptom day from wildfire smoke using an air cleaner: \$141.24	Defensive Behaviour Method based on Richardson <i>et al.</i> (2012) to derive WTP

(Continued)

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Author(s)	Mortality cost estimate	Derivation of mortality cost estimate (in 2019 US dollar values)	Morbidity cost estimates	Derivation of morbidity cost estimates
Kochi <i>et al.</i> (2016)			<i>BenMAP-CE</i> economic values (COI) for ED asthma and HA respiratory (including asthma and pneumonia), minor restricted activity days (MRAD)	Averaging total hospital and ED cost of each patient by county, disease category and age group
			Average medical cost per HA (COI): US\$35 000–83 000	
			Average medical cost per ED (COI): US\$527–722 (COI)	
Jones (2017)			WTP \$442 to avoid 1 day of wildfire smoke	Life satisfaction approach to derive WTP to avoid direct and indirect impacts from wildfires (beyond health) based on changes in subjective wellbeing due to exposure to wildfire smoke
Jones and Berrens (2017)	VSL value of US\$7.48 million	Internally adjusted <i>BenMAP-CE</i> value	WTP \$141.24 for all morbidity impacts	WTP is taken from Jones <i>et al.</i> (2016)
Parthum <i>et al.</i> (2017)			<i>BenMAP-CE</i> COI values for HA respiratory and cardiovascular, by age group	COI estimates are internal to <i>BenMAP-CE</i>
Fann <i>et al.</i> (2018a)	VSL value of US\$10.1 million	Internally adjusted <i>BenMAP-CE</i> value (2010 estimate and 2016 income year)	Local median daily income to determine opportunity cost (i.e. wages lost)	Income from Bureau of Labour statistics
			HA respiratory (COI): \$43 000	EPA estimate for Regulatory Impact Analysis (EPA 2012)
Jones (2018)			WTP for smoke-induced health impacts: \$153	Extension of Jones (2017) using the life satisfaction approach. Differentiation between smoke-induced health impacts and other impacts (lost visibility, amenity)

also applied DBM based on the approach by Richardson *et al.* (2012).

Another way to derive WTP is the life satisfaction approach or subjective wellbeing approach (SWB), which is increasingly applied in various contexts (National Research Council 2014). Jones (2017, 2018) took this approach.

This approach involves taking reported subjective estimates of life satisfaction and explaining them with socioeconomic variables including income level and the variable of interest, here environmental conditions (smoke), geographic location and health status. This approach requires fewer assumptions regarding the rationality of individuals (Levinson 2012) compared with DBM and CVM, and instead considers surveys of SWB as empirical approximations for individual utility (Jones 2018). To isolate the effect of wildfire smoke on SWB, it is necessary to identify locations where smoke plumes coincide with the survey date (or occurred shortly before the survey date). To calculate WTP to avoid a smoke-induced health effect, the marginal rate of substitution (MRS) between income and health status induced by a 1-day increase in smoke exposure is determined. This equals the amount of income compensation required to exactly offset a utility decrement from experiencing 1 additional day of wildfire smoke-induced health effects (Jones 2018).

Jones (2017) and Jones (2018) both use the US Behavioural Risk Factor Surveillance System data (2006–10); however, the latter study tries to isolate the negative impacts on SWB from wildfire smoke due to health-related impacts only (WTP of \$153). Jones' (2017) WTP estimate of \$447 includes all negative impacts on SWB from wildfire smoke (health-related, lost amenity values, changes in ecosystem services, visibility). The differences in results suggest that health costs are a significant portion but only a part of the total cost individuals experience from wildfire smoke exposure.

Few studies so far have estimated the more comprehensive WTP for wildfire morbidity shown in Fig. 1: using DBM, Richardson *et al.* (2012, 2013) estimated US\$100.6 and Jones *et al.* (2016) and Jones and Berrens (2017) estimated US\$141.24 to reduce symptoms by 1 day.

Richardson *et al.* (2013) provide the only CVM WTP estimate, with US\$113.24 and Jones (2017, 2018) determined WTP with the life satisfaction approach as US\$442 and 153 respectively. None of these estimates are endpoint-specific. This is challenging given that, for instance, the disutility from cardiac arrest is likely different from an asthma attack. However, it is noteworthy that all of the measures are relatively close (apart from Jones 2017, as explained above), which can help to

Table 2. Studies since 2006 estimating the economic cost of wildfires: fire assessed, endpoints, use of *Ben-Map* and aggregate cost estimates (where applicable)

All \$ values are 2019 dollars; however, the studies may have used different income adjustments, which likely creates the differences in VSL estimates. COI, Cost of Illness; WTP, Willingness to Pay. HA, hospital admissions; ER, emergency room; ED, emergency department

Author(s)	Event description	Endpoint	<i>BenMAP-CE</i>	Total estimated economic cost
Rittmaster <i>et al.</i> (2006)	Chisholm, Alberta, Canada 2001	Mortality, respiratory and cardiac HAs, ER visits, restricted activity days, asthma symptom days	No	CAN\$18.5 million from an aggregate 1 day increase in PM _{2.5}
Kochi <i>et al.</i> (2012)	Southern California wildfire, 2003	Mortality due to cardiovascular and respiratory illness	No	Additional 133 deaths due to cardiorespiratory illness costing an estimated US\$207.28 million to 2.07 billion
Richardson <i>et al.</i> (2012)	Station Fire, California, 2009	WTP and COI for indirect smoke health impacts	No	WTP for decrease in 1 symptom day from wildfire smoke using an air cleaner: US \$101.57 Private COI based on 1 symptom day: \$11.43 Study did not estimate total cost
Moeltner <i>et al.</i> (2013)	24 large-scale wildfires in Reno/Sparks area Las Vegas, USA, March 2005 to December 2008	Cardiovascular and respiratory morbidity	No	US\$1.3–4.1 million for wildfires in northern Nevada in 2008
Rappold <i>et al.</i> (2014)	Evans Road Fire, rural North Carolina, 2008	Benefits of intervention through adequate smoke forecasting	Yes	Economic benefits of effective interventions: Using <i>BenMAP-CE</i> : US\$58 million (87% of total due to premature mortality) Using HA + ED treatment data: \$473 000 for asthma \$781 000 for congestive heart failure US\$447 000 (non-wildfire-specific DR functions) and US\$540 000 (wildfire-specific DR functions) for ED (asthma), HA admission (all respiratory) and minor restricted activity days
Jones <i>et al.</i> (2016)	Wallow, south-western US wildfire, Albuquerque, 2011	WTP for indirect smoke health impacts	Yes	US\$4.23 million total medical costs (COI)
Kochi <i>et al.</i> (2016)	Southern California wildfire, 2007	Respiratory and acute cardiovascular HA, respiratory and cardiovascular ED visits	No	Derivation of WTP measure to avoid 1 day of wildfire smoke: \$447
Jones (2017)	Impact of smoke from wildfire on life satisfaction using survey data from 2006–10, USA	WTP for total individual damages (health and non-health)	No	
Jones and Berrens (2017)	Western USA wildfires, 2005–15	Mortality, ER all respiratory, HA all respiratory	Yes	\$1.9 billion total health cost or \$177 million per year (2005–15) of which 99% is mortality cost
Parthum <i>et al.</i> (2017)	Benefits of avoiding one wildfire event in a national refuge, Virginia, 2008	Benefits of reducing the occurrence of wildfires (using ED visit data) and COI	Yes	One avoided wildfire catastrophe is worth \$3.69 million or \$306 per hectare burned
Fann <i>et al.</i> (2018a)	Wildfire episodes in the USA 2008–12	Premature mortality, HA	Yes	Between \$13 and \$24 billion per year (2008–12) for short-term exposure Between \$90 and \$154 billion per year (2008–12) from long-term exposure
Jones (2018)	Individual WTP to avoid wildfire smoke exposure in the USA (survey data from 2006 to 2010)	WTP for smoke health impacts (life satisfaction approach)	No	WTP – \$154

contextualise costs across fire and build an evidence base for benefit transfer. In a benefit transfer, estimated values from one or more studies are applied to an unstudied location.

Even though WTP clearly covers a much greater range of the costs, a COI approach is usually much simpler to implement because far fewer variables need to be determined and the data

are more readily available. It is therefore not surprising that most studies looked at COI and very few studies applied WTP for a reduction in health damages; however, WTP values can be up to 30 times greater than the most frequently used COI estimates and five times larger than comprehensive COI estimates (Richardson *et al.* 2013).

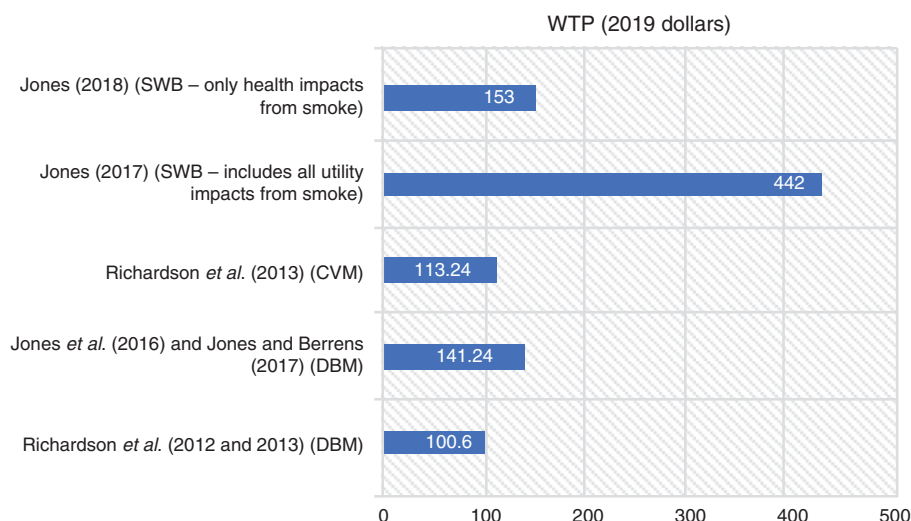


Fig. 1. Willingness to Pay (WTP) estimates per exposed person per day for different studies. SWB, subjective wellbeing approach; CVM, contingent valuation method; DBM, Defensive Behaviour Method.

A way to relate WTP and COI estimates are WTP : COI ratios as recommended and derived by Richardson *et al.* (2013), in other words, multiplying study-specific COI by the ratio to get an estimated WTP. This makes benefit transfer feasible without having to engage in implementing the previously discussed CVM, DBM or life satisfaction approach. However, the chosen WTP : COI ratio needs to be relevant for the particular location or wildfire episode.

Differences in results not only arise from the choice of costing approach but also from the presentation of results: estimates can be in totals or per person, with the former being affected by location (low- v. high-population areas), and whether the study analysed a single wildfire v. multiple fires, or events over multiple years. Table 2 provides an overview of the analysed fires, the considered endpoints, whether *BenMAP-CE* was used and aggregate results for the studies where applicable.

BenMAP-CE

BenMAP-CE is open-source Windows-based software created by the EPA. First developed in 2003 to analyse national-scale air quality policies, the proprietary version was replaced by an open-source tool in December 2013 named *BenMAP-Community Edition*. It allows the user to estimate economic benefits and cost to health due to air quality changes. Its main use has been in the assessment of urban air policy benefits. Studies include both local and regional analyses in the United States but also work elsewhere, mainly in China. The former include work by Fann *et al.* (2011) on air quality policy, health benefits and inequality; by Nowak *et al.* (2013) on the effect of tree removal on PM_{2.5} and associated health effects; and Kheirbek *et al.* (2013) on PM_{2.5} and ozone impacts in New York City, among many others.

Douglass (2008), in a study from the grey literature, was the first to employ *BenMAP-CE* to determine health costs of wildfires in Washington, Idaho and Oregon. Five of the 12 studies on the economic health cost from wildfires have used *BenMAP-CE*

since 2008, a sign of its increasing importance for this type of analysis. Jones *et al.* (2016) link this to its potential to estimate economic impacts of various changes in air quality, whether positive or negative, which until this point ‘was not easy to do in a systematic and controlled way’.

In the case of wildfires, costs are looked at instead of benefits. As a first step, the user imports air quality data for a specific location and population. Then, to estimate the health impacts of the air quality changes, dose–response (DR) functions are used to estimate the incidence rate. In this context, a DR function describes the magnitude of the health impact as a result of a certain increase in the pollutant relative to a baseline incidence. The estimated change in incidences is then multiplied by a given dollar amount using a COI or WTP value. The output can be represented numerically, but *BenMAP-CE* can also be used as a geographic information system (GIS) to map the calculated data. The following sections discuss the key aspects of using *BenMAP-CE* for smoke impacts as a guide to readers who are interested in applying the software for this context.

Air pollution exposure estimates

Most USA studies use freely available air quality data from a variety of EPA monitoring sites, which are located across the country but tend to be concentrated in urban areas. Jones and Berrens (2017), Jones *et al.* (2016) and Moeltner *et al.* (2013) used the readily available AQS (Air Quality Systems) and IMPROVE (Interagency Monitoring of Protected Visual Environments) monitored data (www.epa.gov/outdoor-air-quality-data/interactive-map-air-quality-monitors, accessed 19 August 2020).

AQS is the catch-all term for the ambient air pollution data collected by the EPA, state and local agencies from thousands of monitors (EPA 2019b). In contrast to this, IMPROVE stations only collect air quality data in some of the USA’s protected areas. Therefore, the location of the study will determine the data source used. For example, when looking at urban areas, AQS will be of more use. All studies cited the ease of